

KisMATH: Do LLMs Have Knowledge of Implicit Structures in Mathematical Reasoning

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MAIN FINDING

Mathematical steps in a *chain-of-thought* affect the final answer, and LLMs tend to favour the reasoning paths identified by KisMATH.

OBJECTIVES

- Does *chain-of-thought* imply reasoning?
- Producing intermediate steps is not the same as reasoning. [2]
- Structures are expected from reasoning, e.g., combining axioms or splitting into sub-problems.
- Can we find this structure in LLMs?

APPROACH

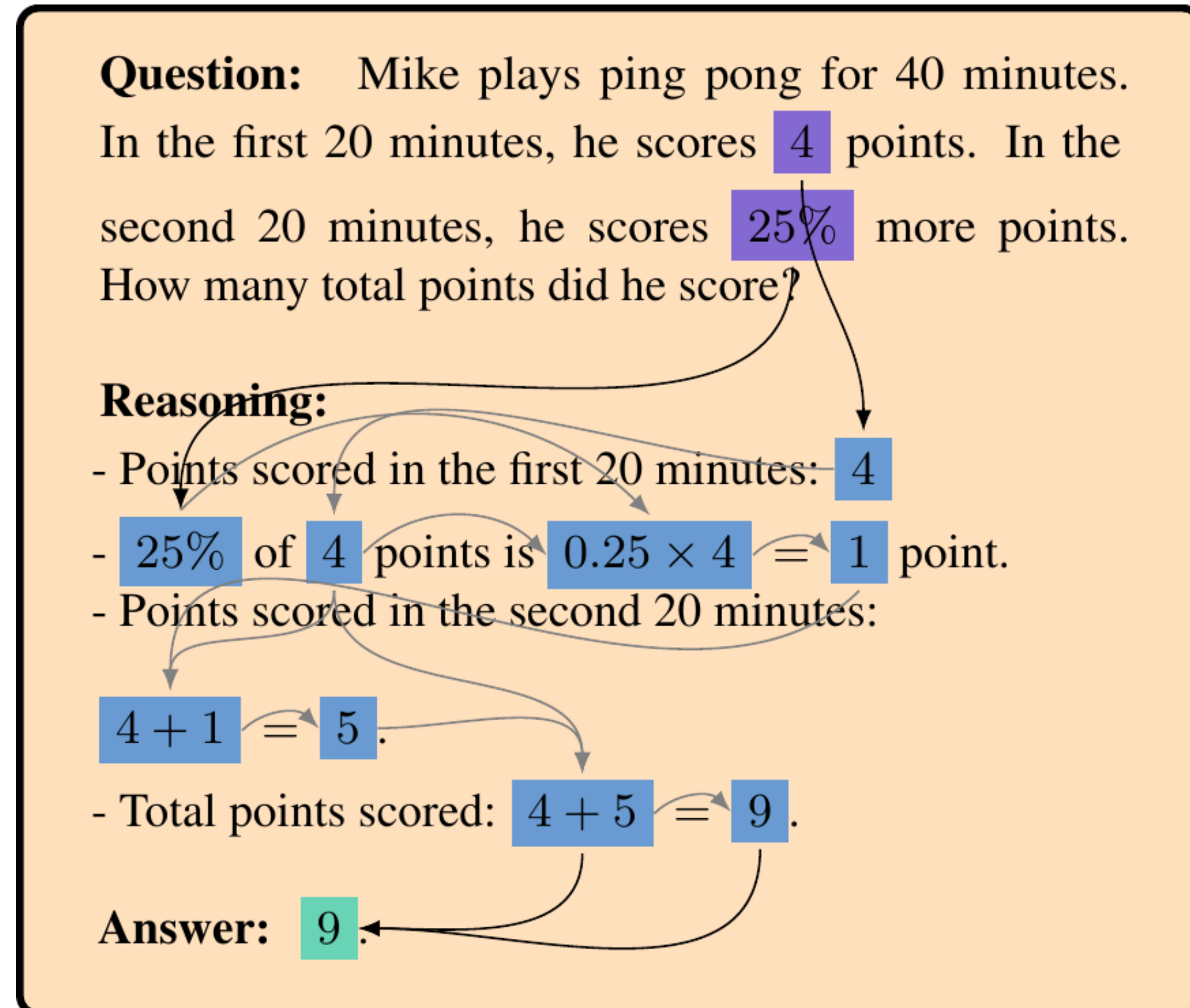


Figure: Example of extracted CCGraph.

- We develop an algorithm to parse mathematical expressions in reasoning traces and their dependencies to *extract the DAG of reasoning—CCGraphs*.
- Using this algorithm, **1671** traces are annotated (GSM8K, MATH500, AIME).
- 15 LLMs at the 1B → 70B parameter scale are analyzed.

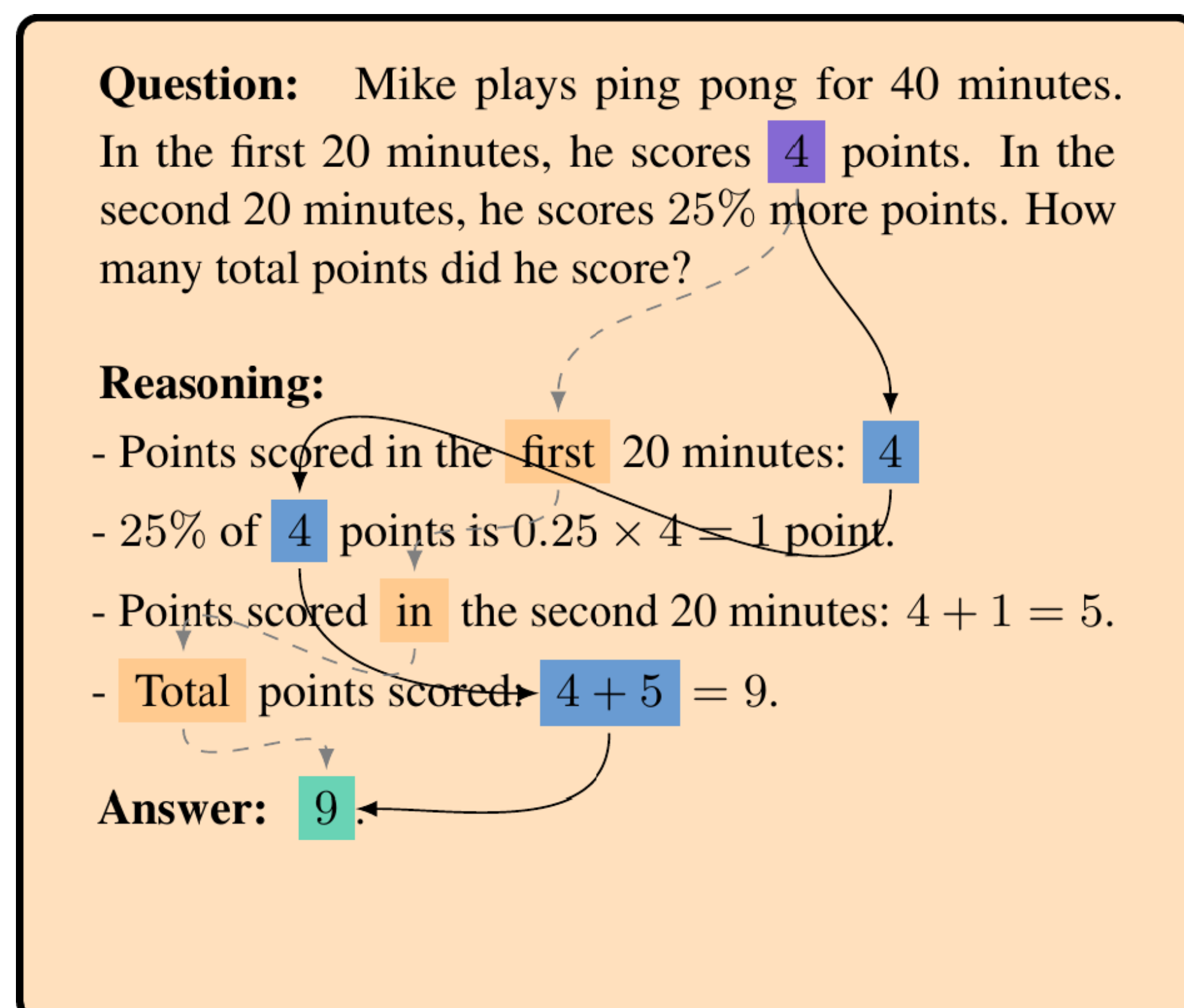


Figure: Example of extracted R path.

R path: Path from a question node to the final answer traversing through the trace. Attention suppression over *intermediate tokens* on this path ensures that these intermediate nodes can't influence other tokens. *Dashed-route* is an equal-length control.

Attribute (avg.)	GSM8K (983)	MATH500 (384)	AIME (304)
$ V $	14.2 ± 4.6	28.7 ± 17.9	50.6 ± 20.9
$ E $	40.8 ± 30.4	260.8 ± 321.2	566.6 ± 487.7
$ \hat{Q} $	3.9 ± 1.4	7.0 ± 4.7	11.1 ± 6.6
$ \hat{R} $	9.3 ± 3.7	20.7 ± 14.8	38.5 ± 18.2
$\text{len}(r)$	6.4 ± 1.8	8.7 ± 2.1	10.9 ± 1.6

Table: Statistics of the KisMATH dataset.

EXPERIMENTS

Intervention: Suppress attention [1] to graph-linked reasoning tokens. *Does the model become less certain of its answers?*

Path preference: Compare the probability of graph-linked R paths with equal-length random controls. *Which path does the model favor?*

RESULTS

Model	GSM8K		
	Orig.	AS	D_{KS}
DeepSeek R1 1.5B	0.02	3.58	1.00
Qwen 3 1.7B	$1e-3$	0.85	0.99
Gemma 3 1B	$2e-3$	0.88	0.98
Llama 3.1 8B	$3e-3$	3.23	0.99
Qwen 2.5 7B	$1e-3$	1.24	0.99
Qwen 2.5 7B Math	0.08	3.30	0.99
DeepSeek R1 8B	0.04	3.69	0.99
DeepSeek R1 8B (0528)	0.01	2.39	0.99
Qwen 3 8B	$8e-4$	0.99	0.99
Gemma 3 12B	$7e-4$	1.02	0.99
DeepSeek R1 32B	$6e-3$	2.73	0.99
Qwen 3 32B	$2e-3$	1.02	0.99
Gemma 3 27B	$4e-4$	0.27	0.99
Llama 3.3 70B	$2e-4$	0.91	0.97
DeepSeek R1 70B	0.02	3.92	0.99

Table: Reasoning nodes are mediators. $H(P_A) \rightarrow$ entropy of the answer. Orig. → Original, AS → Attention Suppressed.

Attention suppression for tokens (e.g., only ~ 14 for GSM8K) in the extracted CCGraph increases answer uncertainty. We can reliably identify a few tokens (based on symbolic parsing) that drastically changes the answer.

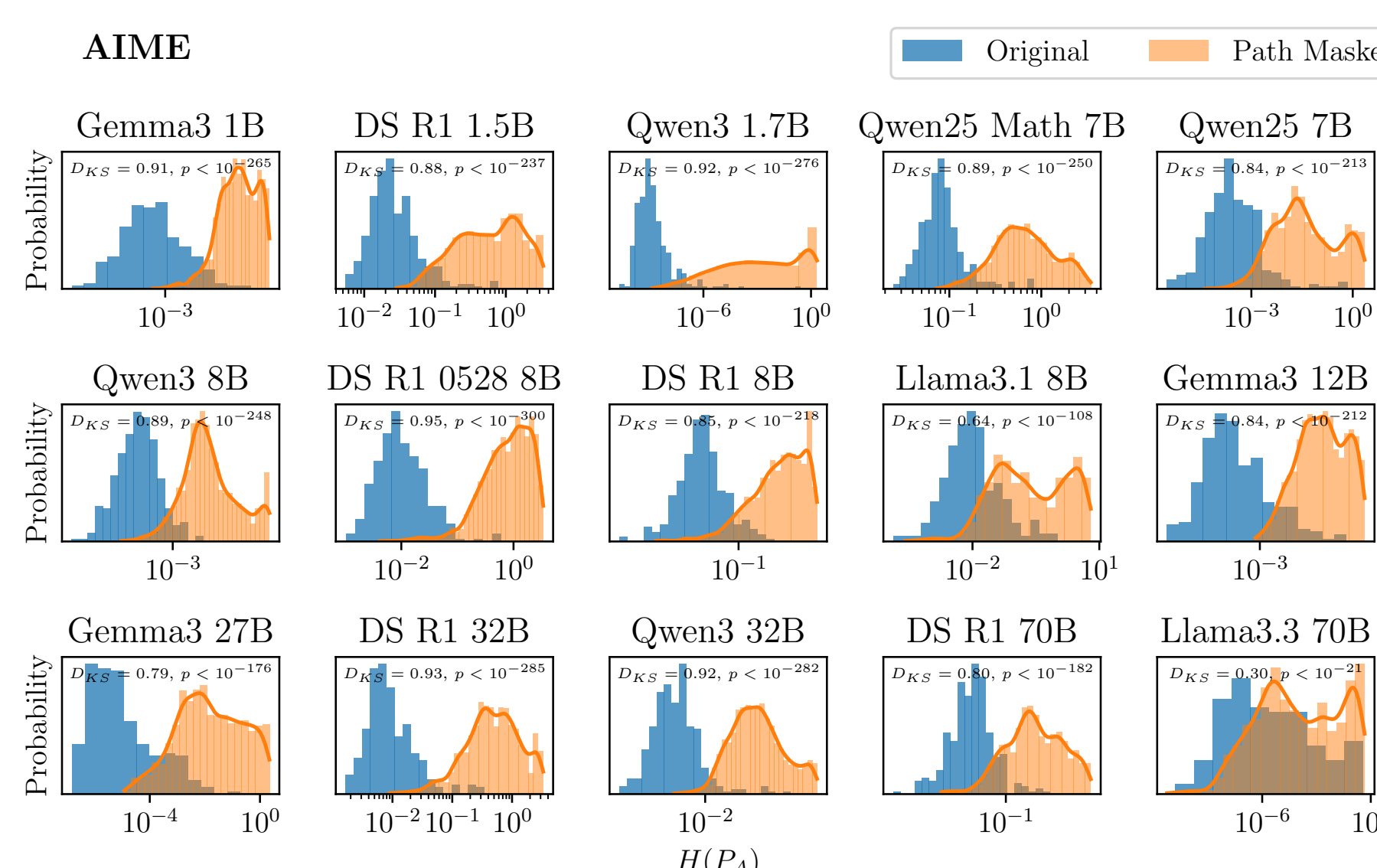


Figure: Path-perturbations affect final answer.

Hiding nodes along an R path makes the answer less certain, the entropy distribution (blue) shows a right-ward shift post-intervention (orange).

We compare the transition probabilities of R paths with respect to M random paths $\tilde{R}_k$, which are not a part of the CCGraph and analyze the distribution of the rank of $P(\mathcal{R})$ (how many random paths have lower transition probabilities than R paths $\mathcal{R}$):

$$\text{rank}_M(\mathcal{R}) = \frac{1}{M} \sum_{k=1}^M \mathbb{I}[P(\mathcal{R}) > P(\tilde{R}_k)]$$

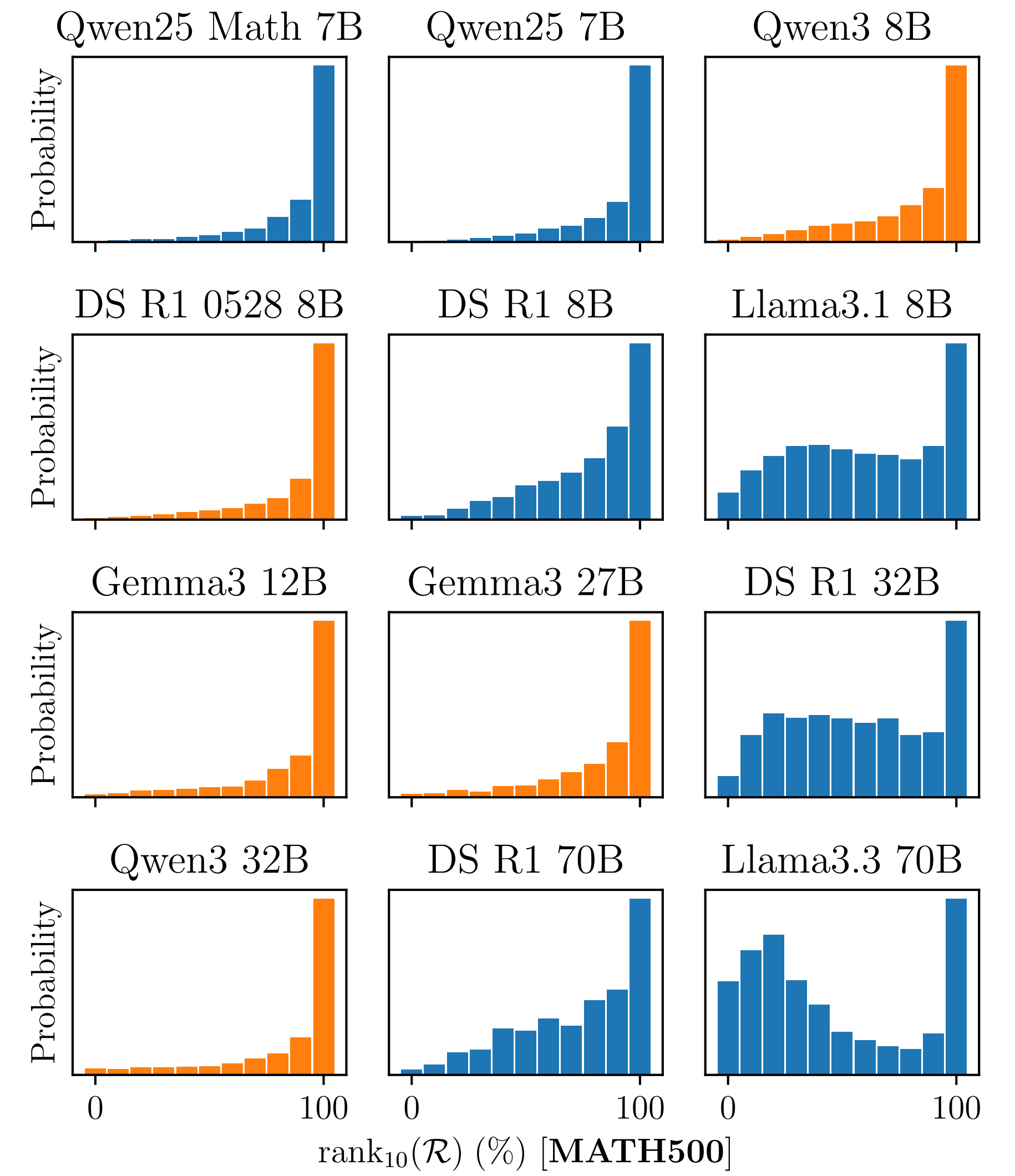


Figure: R paths are preferred.

The pronounced spike at 100 %-ile suggests that LLMs favor R path transitions over other randomly chosen paths.

CONCLUSIONS

- CoT mathematical expressions are not merely decorative. Blocking access to them changes the answer distribution.
- Models show a preference for the R paths extracted by KisMATH.
- These results support a functional role for intermediate steps; they do not prove that every extracted edge is causal. We establish a necessary condition for a reasoning system.

REFERENCES

- [1] Bogdan, P. C., et al. *Thought Anchors: Which LLM Reasoning Steps Matter?* ArXiv preprint, abs/2506.19143, 2025.
- [2] Kambhampati, S., et al. (How) Do reasoning models reason? *Annals of the New York Academy of Sciences*, 1547(1):33–40, 2025. doi:10.1111/nyas.15339.

